

Transaction Costs and the Gains from Trade in Water Markets

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Abstract

Surface water in California is scarce and tightly managed, yet little is traded through a market. This paper studies transaction costs in the existing wholesale water market and estimates the benefits of reducing them. I develop an empirical framework that uses transactions among water districts to analyze welfare, decomposing prices into demand and transaction costs and simulating a market with only physical frictions. Transaction costs are large: median variable costs are 54% of the price, and fixed costs deter many districts from participating. Eliminating them produces gains of several hundred million dollars per year, or 1-3% of statewide water expenditures.

JEL codes D23, F14, R11, Q15, Q25

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1 Introduction

Water supplies are becoming scarcer and more variable in many parts of the world (UNDP 2006). Fueled by population pressure and climate change, water scarcity can increase poverty and conflict and is set to worsen in the coming decades (World Bank Group 2016). To help societies adapt, many observers advocate for greater use of water markets, which – like other markets – could help allocate scarce resources to the most valuable uses. But formal and large-scale water markets are still rare, even in wealthy countries with strong legal institutions (Olmstead 2010). Why? A rich qualitative literature points to unique institutional features of water, complex political economy issues, and transaction costs (Leonard et al. 2019). But relatively little quantitative evidence is available to assess the magnitudes of these barriers and set priorities for reform.

This paper quantifies the importance of transaction costs in California’s wholesale surface water market, the largest in the United States. I ask: How large are the potential gains from reducing transaction costs? In other words, by how much is water misallocated at present, and by how much could it be improved through policy reform? To answer this question, I develop a new empirical framework to estimate demand in the presence of transaction costs that relies on transactions data. The idea is to infer the preferences of water districts – the local government agencies that supply water to farms and households and control most water rights – from their behavior in the market that already exists. After separating observed prices into demand and transaction costs, I simulate a market without transaction costs and estimate the gains from expanded trade.

There is good reason to expect that transaction costs are important in this setting, and that the gains from reducing them could be large. Circumstantial evidence for misallocation is strong: California’s large, diverse economy depends on water supplies that are imported over great distances and quantity-rationed by legacy rights and rules; only a small fraction is traded in the market. Retail prices vary over more than two orders of magnitude: in 2023, commercial and industrial customers in San Diego paid \$2,855 per acre-foot, while just two hours away, agricultural customers in the Imperial Valley paid \$20 per acre-foot.¹ This situation is not due to a lack of infrastructure or

¹An acre-foot, the standard unit of volume for water in the American West, is the amount of

property rights – a comprehensive network of canals, pipelines, rivers, and reservoirs connects nearly all water users in the state, and the legal basis for using and selling water is well-defined – leaving transaction costs as the likely binding constraints.

My starting point is to interpret the high price dispersion and low transaction volume in California's water market as equilibrium outcomes of trade with transaction costs. Transaction costs insert a wedge between observed prices and districts' marginal value of water, and they prevent other trades from happening at all. These costs arise from a variety of sources: internal district decisions, search and contracting, regulatory approval processes, and transportation and delivery. Many transaction costs are unobserved or implicit, so rather than attempting to measure them directly, I infer them from market outcomes.

To do so, I build an empirical model of water districts that trade surface water endowments. The model decomposes observed prices into transaction costs and demand, and each component can be identified using a different source of price variation in the data. Transaction costs are identified by variation across counterparties: when a district makes multiple transactions in the same year, the price gaps reveal differing frictions such as conveyance distance or regulatory reviews. Demand is identified by variation over time: environmental conditions shift endowments from year to year, allowing me to measure how prices respond to quantities. I estimate the model using a comprehensive new dataset on California's water economy, including a nearly complete record of water transaction prices and the universe of wholesale surface water deliveries.

Results show that transaction costs impose significant costs in California's existing water market, reducing the gains from transactions that already occur. Specific regulatory and delivery frictions give rise to large variable transaction costs; for example, buyers pay about 45 percent extra to avoid transactions that involve the federal or state water projects. In total from all sources, median variable transaction costs are 54 percent of the transaction price. And in aggregate, the variable costs of observed transactions are greater than their realized surplus.

Transaction costs also appear to block additional trade that would otherwise bring mutual gains.

water that would cover one acre of land with twelve inches of water. Retail prices were taken from the relevant districts' websites.

Fixed costs are substantial – few districts trade every year, and much of who does trade is explained by frictions such as market inexperience and regulatory reviews – implying that more trade would occur if they were smaller. Marginal valuations remain highly dispersed across districts, implying that increased trade would yield benefits for buyers and sellers.

To quantify the gains from this additional trade, I use the estimated demand model to simulate an efficient market without transaction costs. Because purely physical transportation costs are intrinsic to a water market and can never be eliminated, I build a hydrologic network model of California’s water infrastructure to estimate and include them in the simulation. I estimate that efficient trading would greatly increase trading volumes, and that these new transactions would achieve benefits of \$211 to \$559 million per year. Gains are disproportionately large in dry years, suggesting that water markets can be especially valuable for risk management.

These gains are small relative to water supply expenditures or GDP in California. Part of the explanation is that water is costly to move, so much of the price dispersion is inevitable. There is also reason to suspect that water districts trade more conservatively than their own constituents would prefer. A third possibility is that districts may value the certainty of long-term transactions, which my analysis would miss because it only uses data from short-term leases.

However, the gains are still considerable in absolute terms and may be worth policy reform efforts. Transaction costs from search, negotiation, and contracting could be reduced by new institutions that increase information and coordination, such as a centralized online exchange. Transaction costs from regulatory reviews could be reduced by streamlining existing approval processes.

Two limitations are worth keeping in mind. First, because I study a thin market, the data are sparse; many districts trade infrequently, and others never trade. However, the results are not driven by out-of-sample districts: those that do appear in the trading data account for most of the available water and most of the gains from trade. Second, my simulations generally do not incorporate capacity constraints, but they do not appear to be first-order concerns: in simulations that coarsely

model two major constraints, total gains remain largely unchanged.²

1.1 Related Literature

This paper proposes a new method to estimate demand in water markets that uses transactions data from the existing market. Most prior analyses of the gains from water markets couple detailed engineering models of water infrastructure with mathematical programming models of agricultural production that rely on a large number of calibrated parameters ([Medellín-Azuara et al. 2007](#); [Mérel and Howitt 2014](#)). A recent variation is [Rafey \(2023\)](#), who uses modern production function estimation techniques to value the *ex post* gains from water trading in Australia. My approach differs because it models demand of the relevant actors: water districts, not individual farmers. It is more parsimonious than the programming models and arguably more realistic because it is estimated using observed water market transactions. Even where fine-scale agricultural data is available, working directly with the revealed preferences of water districts makes my approach more policy-relevant so long as districts continue to be the primary market participants.

My approach builds on a large literature in international trade on measuring trade costs, particularly methods that infer costs from price gaps rather than trade flows ([Eaton and Kortum 2002](#); [Atkin and Donaldson 2015](#)). I share the goal of indirectly estimating frictions that are difficult or impossible to measure directly, but I adapt the setup because water is homogeneous, has no production, and transaction costs are bilaterally asymmetric. Many of these features are shared with other settings that involve trade of a factor endowment, such as environmental permit markets ([Stavins 1995](#)) or electricity markets.

I also empirically measure transaction costs in a specific market, which is uncommon outside of international trade. A handful of papers do so in financial markets ([Kyle and Obizhaeva 2016](#)), agricultural labor markets ([Foster and Rosenzweig 2022](#)), supplier contracting ([Boehm 2022](#)), and electricity markets ([Jha and Wolak 2023](#)). A few other papers evaluate the consequences of

²I implicitly assume the water infrastructure forms a fully connected network, which holds for all water districts in my simulation – transfers might require a complicated series of conveyance, exchanges, and/or forbearance but are physically possible ([Hanak and Stryjewski 2012](#)).

reducing transaction costs in financial services (Jack and Suri 2014; de Mel et al. 2022), the upgrade of durable goods (Hodgson 2023), and pollution permit markets (Gangadharan 2000). In water markets, Womble and Hanemann (2020) estimate explicit transaction costs from surveys but miss implicit costs and the gains from forgone transactions, while Carey et al. (2002) and Regnacq et al. (2016) study the effects of transaction costs on trading quantities but not welfare.

Finally, my results shed light on the factors constraining water markets: I show that transaction costs matter, quantify their consequences, and provide evidence on which types are most important. Empirical evidence on water markets has long been limited by both data and methods; I address both gaps. This paper also contributes to a broader literature on the costs of misallocation in markets such as energy (Davis and Kilian 2011), capital (Hsieh and Klenow 2009), labor (Bryan and Morten 2019), and land (Adamopoulos et al. 2022).

2 Background

2.1 California’s water market

California’s water market consists of voluntary sales or leases of the right to use surface water. Transactions are generally legal but lack a central clearinghouse and require approval from one or more regulatory agencies. Moving the water is feasible thanks to one of the world’s most sophisticated systems of water infrastructure.³ Trade occurs only within the state, as legal and infrastructure barriers preclude trade with other states or countries.

The actors in this market are generally water districts rather than farms or households. Water districts are local public agencies that supply water and hold water rights; individual retail customers typically cannot trade outside of their district (Chong and Sunding 2006). Although “water district” as a legal designation applies to only some of the relevant agencies, I use the term more colloquially

³On a shared river or canal, the seller simply takes less water upstream while the buyer takes more downstream. Other transactions are more complex: water may be allowed to flow downstream then pumped into a canal and conveyed hundreds of miles.

to refer to all potential buyers and sellers, including entities such as irrigation districts, cities, county water agencies, and mutual water companies.⁴

Districts obtain water either through (a) surface water rights held with the state or (b) long-term contracts to receive water from the federal and state water projects: California's State Water Project (SWP) and the federal Central Valley Project (CVP) and Lower Colorado River operations. Water deliveries in the projects are rationed using annual allocation percentages, which create year-to-year variation in districts' endowments based on winter precipitation and runoff. More details are described in Online Appendix A.

2.2 Transaction costs

While trading surface water in California is generally legal and feasible, it is costly and difficult. As [Culp et al. \(2014\)](#) write, "A myriad of physical, legal, and regulatory restrictions operate to inhibit the movement of water from one user to another and from one type of use to another." [Western Water Company \(2000\)](#) complains that "The she[e]r complexity of the current water transfer process... is enough to stop most potential transfers from even being proposed."

Economists and legal scholars often describe these difficulties as transaction costs, often pointing to disparities in retail prices or marginal values as evidence of unexploited gains from trade ([Young, 1986](#); [Brewer et al., 2007](#)). Some transaction costs are unavoidable or even efficient ([Colby, 1990](#); [Libecap, 2009](#)), but others persist because of path dependence and distributional conflict ([Edwards and Libecap, 2015](#); [Leonard et al., 2019](#)).

I follow this literature and define transaction costs broadly as all costs of completing a transfer, including both explicit payments and non-monetary frictions.⁵ Transaction costs arise from many

⁴Some farmers are not part of a district and instead hold water rights directly, but they account for a very small fraction of water use in the state.

⁵[Leonard et al. \(2019\)](#) define transaction costs as "the costs of defining, exchanging, and enforcing water rights." The traditional definition focuses on information, search, negotiation, and enforcement ([Coase 1937](#); [Demsetz 1968](#); [Williamson 1979](#); [Barzel 1982](#)); the water markets literature often includes regulatory barriers ([Colby, 1990](#); [Archibald and Renwick, 1998](#); [Bretsen and Hill, 2009](#)) and conveyance costs ([McCann and Easter, 2004](#)). This broader view parallels the

sources; I summarize them here and offer further detail in Online Appendix A.3.

Internal decision costs. To trade water, a district’s board must decide whether to participate, how much to trade, how to acquire water, and whether to proceed with specific transactions. These decisions are costly for several reasons: conflict over how to use the revenues (irrigators who give up water often cannot be directly compensated); uneven distribution of costs and benefits (transfers can harm other irrigators through reduced return flows or shifted tax burdens and harm the broader community by shrinking the local farm economy), and a preference for stability and consensus that leads many boards to avoid controversy ([Bretsen and Hill, 2009](#); [Edwards and Libecap, 2015](#)).

These costs are shaped by governance structure. Districts controlled by landowners may find it easier to sell water than those that respond to a broader electorate, and decision costs are likely higher for districts that answer to more, and more heterogeneous, constituents ([Bretsen and Hill, 2009](#); [Edwards and Libecap, 2015](#); [Bruno et al., 2024](#)). [Hanak \(2003\)](#) observes that “community resistance has soured a number of deals... and has likely prevented others from being proposed.”

Administrative costs. Next, districts must search for a trading partner, negotiate terms, and agree upon a contract. These steps can be costly absent a centralized exchange, standardized contracts, or public data. They create explicit costs – broker fees, attorney fees, staff time – and may carry even larger implicit costs from delays and uncertainty ([Gray et al., 2015](#)). For example, a long-term transfer between the Imperial Irrigation District and the Metropolitan Water District required nine years of negotiations ([Bretsen and Hill, 2009](#)). By contrast, in Australia’s Murray–Darling Basin, online platforms allow deals to clear and deliver within a day ([Johnson, 2015](#)).

Regulatory costs. Then, districts must secure approvals from the State Water Resources Control Board (SWRCB), Department of Water Resources (DWR), the U.S. Bureau of Reclamation (USBR), and/or state and federal wildlife agencies, depending on source, destination, and delivery route. Agencies are required to ensure that transfers do not harm other legal water users, unreasonably

concept of trade costs in international trade.

affect fish, wildlife, or instream uses, or harm the origin county's economy (Water Code §§1725, 1810). The seller must document compliance with relevant laws and analyze effects on other users, the environment, and agency operations. Other parties can protest, requiring negotiations or a public hearing. Approved transfers face ongoing monitoring, including field inspections and even soil sensors ([DWR and SWRCB, 2015](#); [DWR and USBR, 2019](#)).

These reviews are costly and slow.⁶ Explicit costs include agency fees, staff time, attorney fees, and technical consultants; USBR charged \$15,000 to review a 3,675 acre-foot transfer in 2015, and estimates of explicit regulatory costs in Western water markets range from 1% to 112% of price ([Scheer, 2016](#); [Womble and Hanemann, 2020](#)). Implicit costs result from delays and uncertainty. Even short-term transfers take months to approve; long-term transfers can take years ([SWRCB, 1999](#); [Bretsen and Hill, 2009](#)). Third-party protests add risk, as “literally anyone can... either block the transfer or make it too expensive to complete” ([Western Water Company, 2000](#)). Although outright rejections are rare – SWRCB approved 98% of petitions in the 1980s and 90s – proposals are sometimes withdrawn due to local opposition ([Regnacq et al., 2016](#)).

Delivery costs. Finally, the seller must physically move the water to the buyer. This process is costly: a fraction of water is lost along the way, and energy is required to pump water into canals and over mountains. Districts must pay to use conveyance facilities, send more water than is delivered, and can face delivery risks.

Wheeling charges. Many transfers require use of canals and pumps, whose owners typically charge fees that exceed marginal costs to also cover operations and capital costs. These charges can be large enough to double the effective price of water ([Chong and Sunding, 2006](#)).

Conveyance and carriage losses. Agencies assess conveyance losses to account for evaporation and percolation during transport, reducing quantity delivered. They also assess carriage losses of

⁶Regulatory hurdles in particular are not unique to water markets. Similar permitting and approval processes have been shown to impose large costs in sectors such as housing ([Glaeser and Gyourko, 2018](#)), transportation infrastructure ([Brooks and Liscow, 2023](#)), and renewable energy ([Jarvis, 2025](#)).

20-35% for transfers pumped across the Sacramento–San Joaquin Delta to protect ecosystems and other water users from ocean salinity (DWR and USBR, 2019). Both types of losses require the seller to forgo more water than the buyer receives, raising the effective price.

Delta pumping restrictions. Transfers crossing the Delta also face delivery risks under the Endangered Species Act. Transfers are limited to specific months, and pumping may be curtailed unpredictably even during that window, so water released by a seller may never reach the buyer. Timing restrictions can produce an effective surcharge of up to 40%, and even fully approved transfers are not guaranteed to succeed (Hanak and Stryjewski, 2012; DWR and USBR, 2019).

3 Model

I model trade of surface water among water districts in the presence of transaction costs.

3.1 Assumptions

Each district j begins each year t with an endowment of water E_{jt} . There is no production, and water is a homogeneous good.⁷ Districts make bilateral transactions indexed by i .

Assumption 1 (Demand). *Each district has an inverse demand function $V_{jt}(Q_{jt})$, which gives marginal valuation V_{jt} as a function of quantity consumed Q_{jt} . Demand is isoelastic, varies across districts, and may shift over time:*

$$\ln V_{jt} = \eta_j \ln \widetilde{Q}_{jt} + \bar{v}_j + v_{jt} \quad (1)$$

where $\mathbb{E}[v_{jt}] = 0$, and log quantity consumed is centered on the district mean.⁸

⁷Because a buyer receives all water through the same infrastructure, water quality does not vary across potential sellers. It is not literally the same molecules of water that are being sold but rather a series of changes in flows. Few contracts specify water quality (McCann and Easter 2004).

⁸ $\ln \widetilde{Q}_{jt} \equiv \ln Q_{jt} - \ln \overline{Q}_j$, so that the intercept $\bar{v}_j$ represents mean log marginal valuation.

Note this is demand for surface water consumption, not for market transactions. Water districts must have well-defined preferences, but I make no assumptions about how their preferences form. The most substantive restriction is that a district's marginal valuation is constant across transactions within a year. It follows from the plausible assumption that each year's transactions are planned simultaneously, and it is crucial for separating demand from transaction costs. The isoelastic functional form is important for tractability in estimation but not for extrapolation, and in simulations I explore alternative functional forms.

Assumption 2 (Transaction costs). *District j pays a fixed cost f_{jt} to enter the market and complete any transactions. Then, both sales and purchases incur variable transaction costs that are ad valorem (iceberg): District j must give up τ_{jk}^s units of water to sell 1 unit to another district k , and τ_{jk}^b units to buy 1 unit from district k , where $\tau_{jk}^s, \tau_{jk}^b \geq 1$ for all j, k .*

Fixed costs vary across districts and years but are constant across counterparties, a restriction that is tractable and plausible – most fixed costs likely reflect market participation generally, not specific trading partners. Variable costs vary across buyer-seller pairs and are bilaterally asymmetric. Many types of transaction costs are chiefly fixed or variable; others likely load onto both components.⁹ The iceberg specification is uniquely tractable and common in international trade, but unlike that literature, I distinguish between costs paid by the seller and the buyer. Variable costs are constant over time, a key restriction that allows me to separate transaction costs from demand.

Assumption 3 (Components of transaction costs). *Variable transaction costs have observable and unobservable components that are multiplicatively separable: $\ln \tau_{jk}^s = \rho^s \mathbf{B}_{jk} + \check{\tau}_{jk}^s$ (for sales) and $\ln \tau_{jk}^b = \rho^b \mathbf{B}_{jk} + \check{\tau}_{jk}^b$ (for purchases), where $\mathbf{B}_{jk}$ is a vector of observable predictors.*

⁹Some costs scale with transfer volume, such as per-acre-foot wheeling charges and fractional carriage losses. Others are incurred for transfers of any size, such as search costs and attorney fees. Internal decision and regulatory costs are likely both fixed and variable: [DWR and USBR \(2019\)](#) note that review fees depend on transfer size, while survey evidence suggests that explicit legal and regulatory costs increase with volume but less than one-for-one ([Colby, 1990](#); [Scheer, 2016](#); [Womble and Hanemann, 2020](#)).

Since variable transaction costs τ_{jk}^s and τ_{jk}^b are no less than 1, they can be rewritten with a mean-zero error term using a change of variables. For each side of the market $m \in \{s, b\}$:

$$\ln \tau_{jk}^m = \rho^m \mathbf{B}_{jk} + \bar{\tau}_j^m + \tilde{\tau}_{jk}^m \quad (2)$$

where $\bar{\tau}^m \equiv \mathbb{E}[\tilde{\tau}_{jk}^m]$ and $\tilde{\tau}_{jk}^m \equiv \tilde{\tau}_{jk}^m - \bar{\tau}^m$, such that $\mathbb{E}[\tilde{\tau}_{jk}^m] = 0$. The most substantive restriction here is that cost components do not interact – log total costs equals the sum of its parts.

Assumption 4 (No arbitrage). *If a pair of districts trades, prices equalize marginal valuations up to variable transaction costs. For an observed transaction i in year t between any two districts, where the seller is j and the buyer k :*

$$\tau_{jk}^s V_{jt} = p_{ijkt} = \frac{1}{\tau_{kj}^b} V_{kt}. \quad (3)$$

Relative to marginal valuations, the negotiated price gives a premium to the seller, and a discount to the buyer, that is exactly large enough to compensate for the variable transaction costs that each incurs. In equilibrium, each pair of districts that trade equalize their marginal valuations up to variable transaction costs, and each district equalizes its marginal valuation across all districts it trades with. Fixed costs affect who participates in the market but are then sunk, so they do not affect equilibrium prices conditional on the set of participants.

No arbitrage is the crucial assumption underpinning most of the literature on transaction and trade costs.¹⁰ The main reason it might not hold is if there is market power on either side of the market. I investigate this possibility in an appendix to an earlier working paper ([Hagerty 2019](#)) and find little evidence of market power by either buyers or sellers. A separate and contemporaneous analysis by [Tomori et al. \(2024\)](#) finds similar results in the same setting.

¹⁰Online Appendix E.1 discusses possible motivations and derives Equation 3 from a setup involving water brokers.

Not all pairs of districts trade with each other, and not all districts trade at all. Trade will not occur if variable transaction costs are greater than the gap in marginal valuations, or if fixed transaction costs are too large.¹¹ District j enters the market only if the potential surplus from trading makes up for the fixed cost of entry:

$$f_{jt} \leq S_{jt}(q_{jt}, Q_{jt}) \quad (4)$$

where q_{jt} is the potential trading quantity (i.e., net volume purchased in the market if entered), Q_{jt} is total quantity consumed, and potential surplus S_{jt} is the integral of inverse demand over the trading quantity less the sum of prices paid and transaction costs incurred.

3.2 Identification

To take the model to data, I take logs of Equation 3 and allow for a mean-zero residual term, $\mathbb{E}[\epsilon_{ijkt}] = 0$, which might represent errors in measurement or in districts' optimization:

$$\begin{aligned} (\text{Sellers}) \quad \ln p_{ijkt} &= \ln V_{jt} + \ln \tau_{jk}^s + \epsilon_{ijkt} \\ (\text{Buyers}) \quad \ln p_{ijkt} &= \ln V_{jt} - \ln \tau_{jk}^b + \epsilon_{ijkt}. \end{aligned} \quad (5)$$

For a given district, these equations attribute price variation across counterparties to differences in transaction costs, and price variation across years to changes in marginal valuations. Combining equations for buyers and sellers and substituting equations 1 and 2 yields a full model of water market prices:

$$\ln p_{ijkt} = \rho \mathbf{B}_{jk} + \eta_j \widetilde{\ln Q_{jt}} + \bar{v}_j + \bar{\tau}_j + \tilde{\tau}_{jk} + v_{jt} + \epsilon_{ijkt} \quad (6)$$

where $\rho \equiv (1 - b)\rho^s - b\rho^b$, $\bar{\tau}_j \equiv (1 - b)\bar{\tau}_j^s - b\bar{\tau}_j^b$, $\tilde{\tau}_{jk} \equiv (1 - b)\tilde{\tau}_{jk}^s - b\tilde{\tau}_{jk}^b$, and b is an indicator that takes a value of 1 for buyers and 0 for sellers.

¹¹Since the good is homogeneous, the absence of trade does not imply that transaction costs were too large. Each district might trade with a different partner instead, resulting in the sparse trading network seen in the data.

In Equation 6, the demand intercept $\bar{v}_j$ and the mean level of unobserved transaction costs $\bar{\tau}_j$ are not separately identified in transactions data without further restrictions. To identify them, I use a technique inspired by stochastic frontier analysis (Aigner et al. 1977; Kumbhakar et al. 2022).¹²

Assumption 5 (Unobserved transaction costs). *For each district j , trade with at least one counterparty k carries no variable transaction costs: $\min_k \{\tau_{jk}^s\} = 1$ and $\min_k \{\tau_{jk}^b\} = 1$.*

This assumption shifts the distribution of unobserved transaction costs such that overall transaction costs are consistent with Assumption 2. For each side of the market, I can recover

$$\bar{\tau}_j^m = -\min_k \left\{ \rho^m \mathbf{B}_{jk} + \bar{\tau}_{jk}^m \right\}. \quad (7)$$

The mean shifter $\bar{\tau}_j^m$ captures transaction costs uncorrelated with the observable predictors. It creates distance between marginal valuations and equilibrium prices for unobserved costs just as $\rho^m \mathbf{B}_{jk}$ does for observed costs.¹³ The demand intercept $\bar{v}_j$ is then shifted (down for sellers, up for buyers) by the amount of these unobserved transaction costs. Unobserved variable transaction costs are identified only relative to the least costly transaction, so any large variable costs incurred across all transactions will lead to an underestimate of the dispersion in marginal valuations and the potential gains from trade.

4 Estimation

I estimate the full model (Equation 6) by first combining a few terms. For transaction i made by district j with counterparty k in year t , log price p_{ijkt} is explained by observable predictors of transaction costs $\mathbf{B}_{jk}$, log quantity consumed $\ln \widetilde{Q}_{jt}$, a district-specific intercept $\delta_j \equiv \bar{v}_j + \bar{\tau}_j$,

¹²Stochastic frontier analysis was developed to estimate one-sided deviations from an unobserved frontier – typically firm inefficiency relative to a production frontier, but also fitting transaction costs relative to marginal valuations.

¹³The international trade literature (e.g., Helpman et al. 2008) often assumes that unobserved costs are mean-zero distributed around the observable predictors, but here this assumption would likely miss important unobserved costs.

district-pair-specific transaction costs $\tilde{\tau}_{jk}$, and an error term $\varepsilon_{ijkt} \equiv v_{jt} + \epsilon_{ijkt}$:

$$\ln p_{ijkt} = \rho \mathbf{B}_{jk} + \eta_j \ln \widetilde{Q}_{jt} + \delta_j + \tilde{\tau}_{jk} + \varepsilon_{ijkt}. \quad (8)$$

To estimate Equation 8, I propose a four-step procedure that involves sequentially estimating parameters and moving their terms to the left-hand side. This multi-stage procedure allows me to isolate different sources of identifying variation best tailored to estimate each parameter. After estimation, I use Equation 7 to separate δ_j into $\bar{\tau}_j$ and $\bar{v}_j$, giving me all the terms I need to recover transaction costs (Equation 2) and demand (Equation 1).

4.1 Step 1: Variable Transaction Costs from Observable Sources

I estimate observable variable transaction costs ρ incurred by both sellers and buyers by regressing observed prices on a vector of transaction cost predictors $\mathbf{B}_{jk}$:

$$\ln p_{ijkt} = \rho \mathbf{B}_{jk} + \alpha_{jt} + \epsilon_{ijkt}^1. \quad (9)$$

District-by-year fixed effects α_{jt} absorb the other terms from Equation 8, ensuring that coefficients ρ measure price gaps across transactions within district and year. For example, suppose a seller completes transactions with two different buyers in the same year, of which one requires the seller to undergo a particular regulatory review and the other does not. The seller is indifferent between buyers at the margin, so the price gap reveals the equilibrium price premium that compensates the seller for the regulatory review. Note that ρ measures post-contract incidence across buyers and sellers, which may differ from nominal incidence and is otherwise hard to directly observe.¹⁴

¹⁴Some transaction costs are nominally borne by sellers (e.g., regulatory review fees, monitoring and verification), others by buyers (most delivery costs and risks), and many by both (intra-district, administrative, implicit regulatory costs) (DWR and USBR, 2019). However, the actual incidence depends on how contracts are structured. Sellers often invoice buyers for regulatory review fees (Scheer, 2016), and mitigation funds may be included in the transaction price (paid by sellers) or provided separately (paid by buyers) (Hanak, 2003). Details of contract provisions are scarce.

Because the sample of transactions is not large, I avoid overfitting the model by using the least absolute shrinkage and selection operator (lasso) to perform variable selection. I cluster standard errors by district to account for correlation over time, and also by transaction to adjust for the fact that each transaction appears multiple times in the dataset, once for each party.

4.2 Step 2: Market Entry

I estimate the extensive-margin decision of whether to enter the water market in a given year using a probit model. I regress a binary indicator for whether district j completes any transactions in year t on a set of district-level explanatory variables $\mathbf{Z}_{jt}$:

$$\text{Enter}_{jt} = 1 \left(\Pi \mathbf{Z}_{jt} + \epsilon_{jt}^2 > 0 \right). \quad (10)$$

I estimate Equation 10 separately in each year of data (to correct for selection in Step 3) and as a single pooled regression in the full dataset (to understand the overall determinants of market entry). A negative coefficient reveals a source of either fixed transaction costs f_{jt} or of lower potential surplus from trading S_{jt} (Equation 4). Explanatory variables are prior market experience and time-invariant district characteristics (as observable predictors of fixed costs), as well as allocation percentages (as predictors of potential surplus).¹⁵ I use the lasso to select from the full set of predictors, and I cluster standard errors by district.

4.3 Step 3: Demand Elasticity

I estimate the inverse price elasticity of demand η using instrumental variables.¹⁶ First, I adjust observed prices for the variable transaction costs estimated in Step 1. Then, I regress adjusted log

¹⁵I include the full set of candidate instruments from Step 3.

¹⁶In principle, the elasticity may be heterogeneous across districts; in practice, for precision and to avoid weak instrument problems, I estimate a single average elasticity for the market.

prices on log quantity consumed (not traded):

$$\ln p_{ijkt} - \hat{\rho} \mathbf{B}_{jk} = \eta \ln \widetilde{Q}_{jt} + \alpha_j + \phi \hat{\lambda}_{jt} + \theta t + \epsilon_{ijkt}^3. \quad (11)$$

District fixed effects α_j ensure that the elasticity is identified from within-district variation over time.

Instruments. To overcome the joint determination of prices and quantities, I instrument for quantities using administrative changes in annual endowments: allocation percentages in the federal and state water projects. Allocation percentages are determined by precipitation in the mountains during the previous winter, making their year-to-year variation exogenous to demand. They also vary across contract type, as illustrated in Online Appendix Figure A1. Allocation percentages may affect districts directly and others through the market, so I construct many candidate instruments and estimate the model using the IV lasso ([Chernozhukov et al., 2015](#)). Online Appendix B describes instrument construction and the identification assumptions in greater detail.

Sample selection correction. Prices are only observed in years for which a district enters the market, so I correct for sample selection by including inverse Mills ratios $\hat{\lambda}_{jt}$ calculated from year-specific estimates of Equation 10.¹⁷ The primary source of identification for the selection correction is prior market experience, which I assume affect fixed costs but not demand or variable costs, so that they can be excluded from Equation 11.

Time effects. A time trend θt adjusts for changes in demand over time. I omit year fixed effects to avoid spillover bias. Since transaction prices in the same market in the same year are simultaneously determined, within-year comparisons give rise to mechanical spillovers that violate the stable unit treatment value assumption (SUTVA) and eliminate most of the useful variation. Online Appendix E.2 offers a proof of the potentially severe bias from year effects. Although I

¹⁷Bias may result if, for example, a district only enters the market when its endowment is very low, in which case I might only observe prices along a relatively elastic part of its demand curve, resulting in an overestimate of the average price elasticity. This method follows [Wooldridge \(2010\)](#), sections 19.6.2 and 19.9.2.

cannot control for unobserved shocks to demand, the most important time-varying factors relate to water availability, which is flexibly captured by the instruments.

Inference. To account for estimation error in both the outcome variable and the inverse Mills ratios, I calculate standard errors by bootstrapping Steps 1-3 together. I use the Bayesian bootstrap with 1000 iterations and $\exp(1)$ weights blocked by district to account for correlation over time.

4.4 Step 4: District Intercepts and Unobserved Transaction Costs

I estimate district intercepts δ_j and pair-specific unobserved variable transaction costs $\tilde{\tau}_{jk}$ using random effects. I regress log prices (adjusted for both variable transaction costs from Step 1 and demand shifts estimated in Step 3) on time-invariant district characteristics and indicators for both districts and district/counterparty pairs:

$$\ln p_{ijkt} - \hat{\rho} \mathbf{B}_{jk} - \hat{\eta}_j \widetilde{\ln Q_{jt}} = \Gamma \mathbf{X}_j + \mu_j + \tilde{\tau}_{jk} + \varepsilon_{ijkt}. \quad (12)$$

I estimate this three-level random-intercept model via maximum likelihood, with district-pair effects nested within districts.¹⁸ I recover district intercepts by summing fitted values of covariates and predictions of district effects: $\hat{\delta}_j = \hat{\Gamma} \mathbf{X}_j + \hat{\mu}_j$. I again calculate standard errors by bootstrapping Steps 1-4 together, following the process described in Step 3.

5 Data

5.1 Water transactions

No government agency maintains a centralized listing of water transactions in California. Instead, I use a proprietary dataset on water transactions between 1990 and 2015 compiled by WestWater Research, LLC. To my knowledge, this is the most complete dataset of water transactions in

¹⁸I use random effects in this step because they are no longer nuisance parameters – I need to use their estimates in simulations – and fixed effect estimates would be excessively noisy.

California, and it has been used in federal regulatory analyses. It includes many of the same transactions as other datasets previously used in research, but it has a more complete record of prices.¹⁹

I focus my analysis on spot-market transactions (i.e., within-year leases) as opposed to permanent transfers of rights, because their prices can be interpreted without strong assumptions over discount rates, risk premia, and beliefs. Volumes in leases are fixed so per-unit prices are known with certainty; volumes in permanent transfers are stochastic so per-unit prices are uncertain. The data includes 705 spot-market transactions of surface water in which the price is freely negotiated and observed. Online Appendix F describes sample restrictions and data processing in full detail.

5.2 Water quantities and allocation percentages

I assemble a complete accounting of wholesale surface water deliveries and diversions in California, by district, sector, and year, from 1993 through 2015.²⁰ Deliveries, maximum volumes, and allocation percentages in federal and state projects come from DWR and USBR archives; diversions on the basis of water rights come from reports collected by the SWRCB.²¹ I build a crosswalk file to link districts across datasets and a geospatial dataset of water district boundaries. Details of sources, cleaning, and processing of these datasets are described in [Hagerty \(2022\)](#).

5.3 Hydrologic network model

I construct a basic model of California's hydrological network to calculate (a) characteristics of transaction delivery pathways, and (b) purely physical conveyance costs for simulations. The model consists of a set of nodes and edges corresponding to all major water conveyance channels in

¹⁹One dataset was assembled by Gary Libecap ([Brewer et al. 2007](#)), but it lacks many prices and ends in 2008. Another by Ellen Hanak is not publicly available and appears not to focus on prices ([Hanak and Stryjewski 2012](#)).

²⁰Data is available at github.com/hagertynw/data-surface-water and updated through at least 2021.

²¹Allocation percentages are set to 100 percent for water sources outside of the CVP and SWP.

California: rivers, canals, aqueducts, and pipelines. Channels come from the National Hydrography Dataset of the U.S. Geological Survey. Each node and edge is parameterized with physical transportation costs drawn from the literature for energy costs of pumping, conveyance losses, and carriage losses. I run a graph-theory algorithm to obtain the least-cost delivery pathway for each unique pair of planning areas (55 geographic regions defined by the DWR). I assign districts to planning areas based on geography. Online Appendix G describes this model in more detail.

5.4 Observable predictors of transaction costs

I identify and construct predictors of the transaction costs described in Section 2.2.

Variable costs are identified by within-district variation, so their predictors must vary across counterparties. Delivery costs are captured by physical characteristics of delivery routes, which I calculate using the hydrologic network model.²² Regulatory costs are captured by determinants of review count and complexity: whether water is imported into or exported from a federal or state water project, whether the transaction changes the place of use of a post-1914 appropriative water right, and whether the counterparty uses the majority of its water in agriculture.

Fixed costs are identified by between-district variation, so their predictors are district-specific. Administrative costs are captured by trading experience: whether the district completed trades in each of the past three years and the cumulative number of prior years with trades. Regulatory costs are captured by determinants of review requirements: the source of water supplies, and whether agriculture uses the majority of the district's water. Internal decision costs are captured by measures of governance and voting structure, stakeholder number, and stakeholder heterogeneity.²³

²²The variables are elevation gain (pump lift); distance conveyed in rivers; distance conveyed in canals, aqueducts, and pipelines; distance of virtual movement (i.e., transfers against the direction of flow); and whether the transfer crosses the Sacramento–San Joaquin Delta. Elevation gain requires energy use, raising wheeling charges. Distance variables capture conveyance losses (rivers, canals), wheeling charges (canals), and administrative costs that rise with geographic separation (Regnacq et al., 2016). Crossing the Delta introduces carriage losses and delivery risks.

²³Governance: Indicators for district type: water districts, irrigation districts, mutual water companies, and other public agencies, with individual farmers and private companies as the omitted category. Stakeholder number: Log cropland area, as calculated from district shapefiles and USDA's

5.5 Merged dataset and summary statistics

I construct a balanced panel of surface water quantity consumed for all 2,380 districts in California that consume at least 100 acre-feet per year, which represent 98.8% of all surface water consumed. I merge transactions observed for each district, such that each transaction is repeated for each buyer or seller involved. The final dataset has 1,259 transaction-by-district observations, plus 59,165 records of water quantity and market nonparticipation (for years in which districts do not trade and for districts that never trade). Trading districts represent most (63%) of the surface water consumed in California.²⁴

Table 1 shows summary statistics. Panel A shows that the mean transaction price is \$236 per acre-foot (in 2010 dollars) and the distribution of volumes is highly skewed. Panel B shows that most sales come from agriculture, but purchases are shared across sectors. Panel C shows that the most exports are from the Sacramento River hydrologic region and the most imports are from the South Coast and Tulare Lake regions. Figure 1(a) shows that the distribution of prices is highly dispersed even after adjusting for year.

6 Results

6.1 Step 1: Variable Transaction Costs from Observable Sources

Regulatory and delivery frictions indeed appear to be the source of large variable transaction costs (Table 2, column 6).²⁵ Buyers pay approximately 45 percent less for transactions that involve Cropland Data Layer, imputing 0 when a shapefile is unavailable; log maximum volume of water rights or contracts. Heterogeneity: The Herfindahl index (HHI) of shares of water across agriculture and municipal/industrial use.

²⁴Only 7% of districts in the merged dataset ever trade, but most non-trading districts are very small. Because more gains from trade come from larger districts, weighting by surface water consumption is more informative. Online Appendix Figure A3(a) plots districts by water consumption and frequency of appearance in the transactions data.

²⁵Regressors for buyers are multiplied by -1 following Equation 6, so positive coefficients reflect positive transaction costs on both sides of the market: price premiums for sellers and price discounts for buyers.

exporting water from a federal or state water project, and sellers accept approximately 31 percent more for transactions that must cross the Sacramento–San Joaquin Delta.²⁶ Other sources of variable costs selected by the lasso are distance conveyed in rivers, having an agricultural buyer, and review by the State Water Board.

Although not all coefficients are statistically distinguishable from zero, the importance of some of these predictors is supported by the specifications with coarser fixed effects in columns (4)–(5), in which the Delta-crossing coefficient is larger and more precise. Across specifications, all lasso-selected coefficients are positive, supporting the model assumptions and the interpretation of these estimates as variable transaction costs.

Results without variable selection are given in columns (1)–(3). When including all predictors, many coefficients are not statistically distinguishable from zero, and some are negative.²⁷ Still, the overall patterns are broadly consistent across specifications: Buyers require price discounts, and sellers require price premiums, to complete transactions that are subject to these frictions.

6.2 Step 2: Market Entry

Fixed costs appear to be important barriers to trade (Table 3, column 1). Frictions are highly predictive of market entry – just a handful of regressors explain 59 percent of the variation. Market experience is a strong predictor of trading: districts that traded recently are more likely to trade again, which implies that experience reduces fixed costs – and suggests that administrative costs are important.²⁸ Districts in the CVP and SWP are also more likely to trade, likely because many transactions within the projects are streamlined and exempt from SWRCB review – suggesting that

²⁶The finding of transaction costs associated with the Delta is consistent with the previous results of [Regnacq et al. \(2016\)](#), which studies trade frictions in the same setting using a model of trade flows rather than prices.

²⁷Negative coefficients may simply represent reduced costs relative to another predictor. For example, most transactions that involve importing into a project are also subject to State Water Board review, and the sum of those two coefficients is always positive.

²⁸This result also shows that market experience provides independent variation for the sample selection correction.

regulatory costs also matter.

Evidence for internal decision costs is weaker. Districts with more stakeholders (proxied by water right volume) and those governed by broader constituencies (irrigation districts) are more likely to trade. Other measures of governance and stakeholder heterogeneity are insignificant or excluded by the lasso. These results appear contrary to predictions, but positive coefficients can also reflect higher potential surplus, so larger districts may simply have greater gains from trade.

Results also show that potential surplus affects market entry. The coefficient on statewide allocations shows that districts are more likely to trade in dry years, when the gains are large enough to outweigh the fixed costs.

6.3 Step 3: Demand Elasticity

Allocation percentages are strong instruments for both first-stage and reduced-form regressions (Table 3, columns 2 and 3). The F-statistic exceeds conventional thresholds using only two instruments: the district's own allocation percentage and the allocation percentage for all other districts. Coefficients have the expected signs: higher allocations increase quantity consumed and reduce adjusted prices (marginal valuations). Allocations to other districts matter more than own allocations, consistent with the theoretical model: once a district pays the fixed cost to enter the market, quantities and prices are set in statewide equilibrium.

The estimated inverse price elasticity of demand is -1.4 (column 4), implying that a 10% increase in quantity consumed lowers marginal valuations by 14%. The IV lasso selects 3 instruments, and the first-stage F-statistic is now greater because the instruments are selected optimally. Online Appendix Table A2 shows the estimate is robust to more flexible time trends, the set of candidate instruments, dropping the sample selection correction, and using unadjusted prices.

The reciprocal of this estimate implies that districts' price elasticity of demand is -0.69 , conditional on trading. This is the first estimate of the price elasticity of demand for water districts in California's surface water market. Previous studies generally find more inelastic demand for retail

consumers; this comparison suggests that districts are more price-sensitive than their customers.²⁹

6.4 Step 4: District Intercepts and Unobserved Transaction Costs

In the last step of the estimation, a large share of the residual variation is explained by district-level covariates, district-specific effects, and district-pair effects (Online Appendix Table A3). I use these estimates to predict district intercepts δ_j and unobserved variable transaction costs $\tilde{\tau}_{jk}$. I then recover the full demand model and use it to calculate fitted values of marginal valuations for each district in each year.³⁰ Figure 1 plots the distributions of estimated marginal valuations and variable transaction costs, with raw prices for comparison.

Marginal valuations still show considerable dispersion, suggesting that gains are available from reducing transaction costs and increasing trade. Median variable transaction costs are 27 percent on the seller side and 21 percent on the buyer side, for a total of 54 percent of the transaction price.³¹ These costs are somewhat higher for sellers, suggesting asymmetric internal decision or regulatory costs – but buyers also face significant costs, consistent with shared administrative costs or high delivery costs.

7 Simulations

Using the estimated demand model, I simulate California’s water market without transaction costs and calculate the resulting gains from trade. To make computation feasible, I focus on the 154 largest water districts, which together consume 85% of the water in California. I consider three water-supply scenarios: a wet year, a median year, and a dry year. Figure 2 maps fitted marginal valuations for each district in the median-year scenario. Online Appendix C.1 has further details.

²⁹The comparison is inexact. See Online Appendix B.3 for details.

³⁰I recover mean unobserved transaction costs $\bar{\tau}_j$ from $\hat{\tau}_{jk}$ and $\hat{\rho}\mathbf{B}_{jk}$ (Equation 7), district mean marginal valuations $\bar{v}_j$ from $\hat{\delta}_j$ and $\hat{\tau}_j$ (Equation 8), and inverse demand from $\hat{v}_j$ and $\hat{\eta}$ (Equation 1).

³¹Values do not sum due to log transformations. Mean values are different between sellers and buyers in a t -test.

7.1 Surplus from the existing market

I first calculate the economic surplus achieved to date by observed transactions. Figure 3, Panel (a) maps the geographic patterns of these transactions. Sellers tend to be in the Sacramento Valley and northern San Joaquin Valley; buyers tend to be in urban coastal areas and the southern San Joaquin Valley. Table 4, Panel A, reports that the total volume of water traded ranges between 35,000 and 289,000 acre-feet per year depending on water conditions.

Surplus for each district S_j (suppressing time subscripts) is the difference in marginal valuations before and after trading, integrated over the quantity traded q_j (i.e., net purchases) and evaluated using the estimated demand model (Equation 1):

$$S(q_j, Q_j) = \int_{Q_j - q_j}^{Q_j} [V_j(\varphi) - V_j(Q_j)] d\varphi. \quad (13)$$

Table 4, Panel A reports that observed transactions result in economic surplus of \$8.3 million per year in a median year. In a dry year, total gains are larger but still relatively small, at \$55.2 million per year.

7.2 Counterfactual simulations

To simulate an efficient market, I solve the social planner's problem in a constrained optimization problem. Because an ideal market could implement the efficient allocation, the increase in surplus represents the potential gains from trade. Because purely physical transportation costs are unavoidable, I include conveyance costs from my hydrologic network model as costs in the objective function.³² I do not model constraints on flow capacity except in two specific scenarios.

The social planner chooses the vector of bilateral transaction quantities q_{jk} (net volumes delivered from district j to district k , for all unique pairs $k > j$) that maximizes the aggregate value of water net of conveyance costs, subject to a resource constraint. Aggregate value is the area under

³²Online Appendix Figure A2 illustrates these costs. I calibrate conveyance costs instead of empirically estimating them because they are correlated with other sources of transaction costs and therefore difficult to isolate.

each district's demand curve between its observed quantity Q_j and simulated final quantity Q_j^f , summed over districts. Conveyance costs are per-unit and directionally asymmetric. The resource constraint requires that final quantities be nonnegative. Together, the full optimization problem is:

$$\max_{\{q_{jk}\}_{j,k>j}} \sum_j \int_{Q_j}^{Q_j^f} V_j(\varphi) d\varphi - \sum_j \sum_{k>j} \left[1(q_{jk} > 0) c_{jk} - 1(q_{jk} < 0) c_{kj} \right] q_{jk} \quad (14)$$

subject to

$$\begin{aligned} \text{(definition of final quantities)} \quad Q_j^f &= Q_j - \sum_{k>j} q_{jk} + \sum_{k<j} q_{kj} \quad \forall j \\ \text{(resource constraint)} \quad Q_j^f &\geq 0 \quad \forall j. \end{aligned}$$

I solve this problem using the `patternsearch` solver in Matlab, using observed quantities as the initial conditions. Online Appendix E.3 proves that the solution to this planner's problem also satisfies the conditions of an efficient market. Efficient trading creates two types of benefits: reduced costs of transactions that already take place, and gains from new transactions that would not otherwise take place. I calculate both types of benefits and sum across districts.³³

7.3 Gains from an efficient market

Eliminating transaction costs – other than the physical costs of conveyance – would increase surface water trade dramatically, by 3 to 4 million acre-feet per year (Table 4, Panel B, Scenario 2). Geographic patterns of this simulated trade are mapped in Figure 3 (and Figure A4 in color). In a median year (Panel (c)), large quantities of water are sold by major agricultural regions and purchased by urban coastal areas and the southern San Joaquin Valley, which is also agricultural but

³³Gains from new transactions are calculated in the same way as Equation 13, evaluated between Q_j and Q_j^f . Online Appendix E.4 proves that the sum of these gains across districts is equal to the objective in Equation 14. Gains from reduced costs of observed transactions are $R_j \equiv (V_j(Q_j) - V_j(Q_j^f)) q_j$. I restrict R_j to be non-negative, since negative values occasionally arise if the simulation predicts net trading in the opposite direction of observed trading.

particularly water-scarce. These patterns intensify in a dry year (Panel (d)). The most purchases are by the Los Angeles area, where water shortages are costliest (Buck et al. 2016).

These new transactions would generate gains of \$211 million to \$559 million, as valued by the water districts. In addition, districts would avoid \$5 to \$85 million of transaction costs in pre-existing transactions. These estimates dwarf the surplus generated by the existing market; transaction costs alone exceed the realized surplus of observed transactions. Gains are considerably larger in dry years, suggesting that water markets are most valuable in California when water is scarcest, and that short-term leases offer valuable flexibility for managing risk. But in broader context, the gains from trade represent just 1-3% of California's annual water supply expenditures.³⁴

7.4 Extensions and sensitivity checks

With fixed transaction costs. Scenario 3 simulates a market that keeps the fixed costs of market entry but continues to eliminate variable costs (other than conveyance costs). I implement this scenario by restricting trade to districts predicted to enter the market by the model estimated in Step 2.³⁵ Variable costs alone are quantitatively important: Relative to the status quo, gains are \$77 to \$393 million per year. But fixed costs are also important: Relative to the efficient scenario that eliminates both, gains are lower by \$81 to \$166 million per year. These results also suggest that much of the potential gains from trade can be attained by relatively few market participants and relatively little new transaction volume.

Without conveyance costs. I also investigate the consequences of purely physical conveyance costs. How much more trade would occur if water could be traded costlessly? Scenario 4 simulates an efficient market with all transaction costs c_{kl} set to zero, implausibly. More trading occurs, especially in non-dry years. Relative to Scenario 2, gains are about double in a dry year, at \$1.1 billion, and triple in median and wet years. These results suggest that much of the observed

³⁴Hanak et al. (2014) estimate that federal, state, and local agencies spent \$16.9 billion on water supply, and \$30.5 billion on all water-related spending, per year between 2008 and 2011.

³⁵Only 12 districts enter in the wet year, 28 in the median year, and 43 in the dry year.

dispersion in water valuations in different parts of California is unavoidable and due to the fact that water is costly to move.

Environmental and infrastructure constraints. Because my main scenario does not model hydrological capacity constraints, it is natural to ask how important they might be. I consider the most important juncture in California’s hydrology, the Sacramento–San Joaquin Delta. Increasing flows through the Delta may be undesirable due to constraints to protect water quality and endangered species. How much would maintaining these constraints affect the gains from trade? There is reason to expect the answer to be large: Step 1 showed the Delta is associated with large variable transaction costs, and the new trade volume moving across the Delta in Scenario 2 exceeds the capacity of the state’s proposed Delta Conveyance Project.³⁶

Scenario 5 imposes a hard constraint at the Delta, by simulating separate markets north and south of the Delta and then combining results. Trading volumes and economic gains are lower than in Scenario 2, but the difference is small. The lost trade across the Delta is replaced by an near-equal volume of trade within regions (i.e., north and south of the Delta). The gains are relatively unaffected because marginal valuations have greater variation within region than between them. Most gains from trade come from collapsing this variation within each region rather than joining the two regions.³⁷ This result suggests that most of the gains from water markets could be achieved without relaxing environmental regulations or building major new infrastructure.

Online Appendix C.2 describes more sensitivity checks; the main results are qualitatively robust.

7.5 Fixed costs

As a final exercise, I quantify fixed costs using bounds implied by Equation 4. For tractability, I now assume fixed costs are constant in time. For each district observed trading, the fixed cost must be no greater than surplus attained, so an upper bound is given by the minimum nonzero surplus

³⁶Estimates suggest this project would enable 0.1 to 1.0 million acre-feet per year to be transferred across the Delta (DWR, 2025), as compared with 0.9 to 1.8 million in Scenario 2.

³⁷Online Appendix Figure A5 provides a visualization.

observed in any year. For each district never observed trading, the fixed cost must be greater than forgone surplus, so a lower bound is given by the maximum surplus attained in any year of the efficient-trading simulation (Scenario 2). I then assume these bounds hold with equality.³⁸

Figure 4 plots the distribution of estimated fixed costs across districts; the median is \$55,600 and the mean is \$973,000. As expected, fixed costs are greater for districts that never trade. The two methods produce overlapping densities, consistent with the idea that they estimate different parts of the same underlying distribution. Aggregate fixed costs are \$221 million per year, suggesting that between 40 and 100 percent of simulated gains are attributable to fixed costs, depending on the year. This result is broadly consistent with the indirect method of assessing fixed costs in Scenario 3, confirming that both fixed and variable costs are important in preventing gains from trade.

8 Discussion and Conclusion

I estimate that the gains available from reducing transaction costs in California's water market, including both direct cost savings and the gains from trade, are \$225 to \$644 million per year. These gains represent the potential benefits of legal and policy reforms that improve the allocative efficiency of water use across water districts.

Are the potential gains large or small? I suggest they are moderate. On one hand, annual trading volumes would increase by two to three orders of magnitude, and the dollar value of benefits is considerable, so if the costs of achieving them are low, they are worth pursuing. On the other hand, only 10 to 15 percent of California's water consumption would change hands, and the gains from trade are less than 2 percent of California's agricultural GDP and 4 percent of the state's water supply expenditures (Hanak et al. 2014).

Why are the potential gains smaller than anecdotal evidence suggests? Several reasons may contribute. First, water is costly to move: Gains are 2-3 times greater when I simulate trading with

³⁸This is a plausible approximation for observed traders, few of whom trade every year, so if fixed costs were lower, they would trade more often. It is likely more of an underestimate for never-traders.

costless conveyance, suggesting that much of the differences in water valuations across California are the inevitable result of physics. Second, my analysis may miss some forms of transaction costs. In particular, unobserved internal decision costs may lead water districts to trade less than their constituent farmers or households would prefer (Libecap 2009). Third, by analyzing the lease market, I may understate the potential gains from longer-term transactions. Urban districts and perennial crop growers often place a high value on water supply certainty, so they may value one-year leases in the data less than the annualized value of a permanent water right (or than leases in a thicker and more predictable market).³⁹

Which types of transaction costs matter most, and how can they be reduced? Delivery costs are substantial but cannot be reduced by much, because they appear to be fully explained by unavoidable physical conveyance costs. In particular, I estimate large variable costs of crossing the Sacramento–San Joaquin Delta. But once the simulations account for modeled conveyance costs, imposing additional pumping constraints has little further effect on the gains from trade. This suggests that trade across the Delta is limited primarily by physical costs (i.e., water losses that are necessary to maintain the hydrological system) rather than regulations that protect endangered species, at least for short-term leases.⁴⁰

In contrast, regulatory and administrative costs – also important barriers to trade – are more amenable to reform. Regulatory reviews are associated with large transaction costs, both variable

³⁹Evidence for this point, as suggested by a referee, is seen in the tendency of some urban water districts to invest in desalination or water recycling plants. The levelized cost of water from these methods is typically \$1,000 to 3,000 per acre-foot (Cooley et al. 2019), well above the mean marginal valuations I estimate for urban districts.

⁴⁰Environmental pumping restrictions may still be costly in longer-term transactions due to delivery risks. Transfers that cross the Delta have a higher lease-to-sale price ratio (0.48) than do transfers within the south-of-Delta region (0.15). This comparison – based on Grainger and Costello (2014) and suggested by a referee – suggests that long-term risk lowers the value of a permanent right relative to a lease. Reducing these risks would require either (a) loosening pumping restrictions by reinterpreting Endangered Species Act protections for fish in the Delta, or (b) building new conveyance capacity, such as the state’s proposed Delta Conveyance Project, a \$20 billion, 45-mile tunnel (DWR, 2025). Both options bring costs that must be weighed against the potential gains.

and fixed. Administrative costs are implied by patterns of fixed costs – market experience predicts market participation, and participation remains low even for districts subject to fewer regulatory reviews. Administrative costs can be remedied with better information: The state could create a new centralized online exchange to match parties and execute transfers, as in Australia, or at least begin to routinely collect and publish prices and other data about water transfers (Gray et al., 2015). Internal decision costs can be reduced by creating new legal pathways for individual farmers to unilaterally transfer unused water outside their district (Culp et al., 2014).

Regulatory costs can be reduced by streamlining existing review processes in three ways. First, the number of reviews can be minimized by combining processes across agencies and exempting transfers that are small, temporary, or within the same watershed or project (Sellers et al., 2016). Second, the processes can be simplified by standardizing methods and pre-calculating values used to quantify water consumption and its externalities (Gray et al., 2015). Finally, uncertainty can be reduced by replacing public reviews with staff-level technical determinations (Hanak and Stryjewski, 2012). These proposals mirror those for reducing regulatory costs in other sectors, such as housing and infrastructure construction.

References

- Adamopoulos, Tasso, Loren Brandt, Jessica Leight, and Diego Restuccia, “Misallocation, Selection, and Productivity: A Quantitative Analysis With Panel Data From China,” *Econometrica*, 2022, 90 (3), 1261–1282.
- Aigner, Dennis, C. A. Knox Lovell, and Peter Schmidt, “Formulation and estimation of stochastic frontier production function models,” *Journal of Econometrics*, July 1977, 6 (1), 21–37.
- Archibald, Sandra O. and Mary E. Renwick, “Expected Transaction Costs and Incentives for Water Market Development,” in K. William Easter, Mark W. Rosegrant, and Ariel Dinar, eds., *Markets for Water: Potential and Performance*, Boston, MA: Springer US, 1998, pp. 95–117.
- Atkin, David and Dave Donaldson, “Who’s Getting Globalized? The Size and Implications of

- Intra-national Trade Costs,” Working Paper 21439, National Bureau of Economic Research July 2015.
- Barzel, Yoram**, “Measurement Cost and the Organization of Markets,” *The Journal of Law & Economics*, 1982, 25 (1), 27–48.
- Boehm, Johannes**, “The Impact of Contract Enforcement Costs on Value Chains and Aggregate Productivity,” *The Review of Economics and Statistics*, 01 2022, 104 (1), 34–50.
- Bretsen, Stephen N. and Peter J. Hill**, “Water Markets as a Tragedy of the Anticommons,” *William & Mary Environmental Law and Policy Review*, 2009, 33 (3), 723–783.
- Brewer, Jedidiah, Robert Glennon, Alan Ker, and Gary Libecap**, “Transferring Water in the American West: 1987–2005,” *University of Michigan Journal of Law Reform*, 2007, 40 (4), 1021–1053.
- Brooks, Leah and Zachary Liscow**, “Infrastructure Costs,” *American Economic Journal: Applied Economics*, April 2023, 15 (2), 1–30.
- Bruno, Ellen M., Nick Hagerty, and Arthur R. Wardle**, “The Political Economy of Groundwater Management: Descriptive Evidence from California,” in Gary D. Libecap and Ariel Dinar, eds., *American Agriculture, Water Resources, and Climate Change*, National Bureau of Economic Research: University of Chicago Press, 2024, pp. 343–365.
- Bryan, Gharad and Melanie Morten**, “The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia,” *Journal of Political Economy*, 2019, 127 (5), 2229–2268.
- Buck, Steven, Maximilian Auffhammer, Stephen Hamilton, and David Sunding**, “Measuring Welfare Losses from Urban Water Supply Disruptions,” *Journal of the Association of Environmental and Resource Economists*, 2016, 3 (3), 743–778.
- Carey, Janis, David L. Sunding, and David Zilberman**, “Transaction costs and trading behavior in an immature water market,” *Environment and Development Economics*, 2002, 7 (04), 733–750.
- Chernozhukov, Victor, Christian Hansen, and Martin Spindler**, “Post-Selection and Post-Regularization Inference in Linear Models with Many Controls and Instruments,” *American Economic Review: Papers & Proceedings*, 2015, 105 (5), 486–490.

- Chong, Howard and David Sunding**, “Water Markets and Trading,” *Annual Review of Environment and Resources*, 2006, 31, 239–264.
- Coase, R. H.**, “The Nature of the Firm,” *Economica*, November 1937, 4 (16), 386–405.
- Colby, Bonnie G.**, “Transactions Costs and Efficiency in Western Water Allocation,” *American Journal of Agricultural Economics*, 1990, 72 (5), 1184–1192.
- Cooley, Heather, Rapichan Phurisamban, and Peter Gleick**, “The cost of alternative urban water supply and efficiency options in California,” *Environmental Research Communications*, May 2019, 1 (4), 042001.
- Culp, Peter W., Robert Glennon, and Gary Libecap**, “Shopping for Water: How the Market Can Mitigate Water Shortages in the American West,” Discussion Paper, The Hamilton Project & Stanford Woods Institute for the Environment October 2014.
- Davis, Lucas W. and Lutz Kilian**, “The Allocative Cost of Price Ceilings in the U.S. Residential Market for Natural Gas,” *Journal of Political Economy*, 2011, 119 (2), 212–241.
- de Mel, Suresh, Craig McIntosh, Ketki Sheth, and Christopher Woodruff**, “Can Mobile-Linked Bank Accounts Bolster Savings? Evidence from a Randomized Controlled Trial in Sri Lanka,” *The Review of Economics and Statistics*, 03 2022, 104 (2), 306–320.
- Demsetz, Harold**, “The Cost of Transacting,” *The Quarterly Journal of Economics*, 1968, 82 (1), 33–53.
- DWR**, “The Delta Conveyance Project in Practice,” Technical Report, California Department of Water Resources 2025.
- **and SWRCB**, “Background and Recent History of Water Transfers in California,” Technical Report, California Department of Water Resources & State Water Resources Control Board July 2015.
- **and USBR**, “Draft Technical Information for Preparing Water Transfer Proposals (Water Transfer White Paper),” Technical Report, California Department of Water Resources & U.S. Bureau of Reclamation, California–Great Basin Region December 2019.
- Eaton, Jonathan and Samuel Kortum**, “Technology, Geography, and Trade,” *Econometrica*,

2002, 70 (5), 1741–1779.

Edwards, Eric C. and Gary D. Libecap, “Water institutions and the law of one price,” in “Handbook on the Economics of Natural Resources,” Edward Elgar Publishing, March 2015, pp. 442–473.

Foster, Andrew D. and Mark R. Rosenzweig, “Are There Too Many Farms in the World? Labor Market Transaction Costs, Machine Capacities, and Optimal Farm Size,” *Journal of Political Economy*, 2022, 130 (3), 636–680.

Gangadharan, Lata, “Transaction Costs in Pollution Markets: An Empirical Study,” *Land Economics*, 2000, 76 (4), 601–614.

Glaeser, Edward and Joseph Gyourko, “The Economic Implications of Housing Supply,” *Journal of Economic Perspectives*, February 2018, 32 (1), 3–30.

Grainger, Corbett A. and Christopher J. Costello, “Capitalizing property rights insecurity in natural resource assets,” *Journal of Environmental Economics and Management*, 2014, 67 (2), 224–240.

Gray, Brian, Ellen Hanak, Richard Frank, Richard Howitt, Jay Lund, Leon Szeptycki, and Barton "Buzz" Thompson, “Allocating California’s Water: Directions for Reform,” Technical Report, Public Policy Institute of California Water Policy Center November 2015.

Hagerty, Nick, “Liquid Constrained in California: Estimating the Potential Gains from Water Markets,” October 2019. Working paper.

—, “Adaptation to Surface Water Scarcity in Irrigated Agriculture,” 2022. Working paper.

Hanak, Ellen, *Who Should Be Allowed to Sell Water in California? Third-Party Issues and the Water Market* 2003.

— **and Elizabeth Stryjewski**, “California’s Water Market, By the Numbers: Update 2012,” PPIC Report, Public Policy Institute of California November 2012.

—, **Brian Gray, Jay Lund, David Mitchell, Caitrin Chappelle, Andrew Fahlund, Katrina Jessoe, Josué Medellín-Azuara, Dean Mischynski, James Nachbaur, and Robyn Suddeth**, “Paying for Water in California,” PPIC Report, Public Policy Institute of California March 2014.

- Helpman, Elhanan, Marc Melitz, and Yona Rubinstein**, “Estimating Trade Flows: Trading Partners and Trading Volumes,” *Quarterly Journal of Economics*, May 2008, 123 (2), 441–487.
- Hodgson, Charles**, “Trade-ins and Transaction Costs in the Market for Used Business Jets,” *American Economic Journal: Microeconomics*, November 2023, 15 (4), 350–91.
- Hsieh, Chang-Tai and Peter J. Klenow**, “Misallocation and Manufacturing TFP in China and India,” *Quarterly Journal of Economics*, 2009, 124 (4).
- Jack, William and Tavneet Suri**, “Risk Sharing and Transactions Costs: Evidence from Kenya’s Mobile Money Revolution,” *American Economic Review*, January 2014, 104 (1), 183–223.
- Jarvis, Stephen**, “The Economic Costs of NIMBYism: Evidence from Renewable Energy Projects,” *Journal of the Association of Environmental and Resource Economists*, 2025, 12 (4), 983–1022.
- Jha, Akshaya and Frank A. Wolak**, “Can Forward Commodity Markets Improve Spot Market Performance? Evidence from Wholesale Electricity,” *American Economic Journal: Economic Policy*, May 2023, 15 (2), 292–330.
- Johnson, Nathanael**, “California Has a Real Water Market — But It’s Not Exactly Liquid,” *Grist*, May 2015.
- Kumbhakar, Subal C., Christopher F. Parmeter, and Valentin Zelenyuk**, “Stochastic Frontier Analysis: Foundations and Advances II,” in Subhash C. Ray, Robert Chambers, and Subal Kumbhakar, eds., *Handbook of Production Economics*, Singapore: Springer, 2022, pp. 1–38.
- Kyle, Albert S. and Anna A. Obizhaeva**, “Market Microstructure Invariance: Empirical Hypotheses,” *Econometrica*, 2016, 84 (4), 1345–1404.
- Leonard, Bryan, Christopher Costello, and Gary D. Libecap**, “Expanding Water Markets in the Western United States: Barriers and Lessons from Other Natural Resource Markets,” *Review of Environmental Economics and Policy*, 2019, 13 (1), 43–61.
- Libecap, Gary D.**, “Chinatown Revisited: Owens Valley and Los Angeles—Bargaining Costs and Fairness Perceptions of the First Major Water Rights Exchange,” *The Journal of Law, Economics, and Organization*, October 2009, 25 (2), 311–338.

- McCann, Laura and K. William Easter**, “A framework for estimating the transaction costs of alternative mechanisms for water exchange and allocation,” *Water Resources Research*, September 2004, 40 (9), 2003WR002830.
- Medellín-Azuara, Josué, Julien J. Harou, Marcelo A. Olivares, Kaveh Madani, Jay R. Lund, Richard E. Howitt, Stacy K. Tanaka, Marion W. Jenkins, and Tingju Zhu**, “Adaptability and adaptations of California’s water supply system to dry climate warming,” *Climatic Change*, 2007, 87.
- Mérel, Pierre and Richard Howitt**, “Theory and Application of Positive Mathematical Programming in Agriculture and the Environment,” *Annual Review of Resource Economics*, 2014, 6 (1), 451–470.
- Olmstead, Sheila M.**, “The Economics of Managing Scarce Water Resources,” *Review of Environmental Economics and Policy*, 2010, 4 (2), 179–198.
- Rafey, Will**, “Droughts, Deluges, and (River) Diversions: Valuing Market-Based Water Reallocation,” *American Economic Review*, 2023, 113 (2), 430–471.
- Regnacq, Charles, Ariel Dinar, and Ellen Hanak**, “The Gravity of Water: Water Trade Frictions in California,” *American Journal of Agricultural Economics*, 2016, 98 (5), 1273–1294.
- Scheer, Jennifer B.**, “Participation, Price, and Competition in the Sacramento Valley Water Market,” Master’s thesis, University of California, Davis 2016.
- Sellers, Scott, Matthew Zaragoza-Watkins, Christina Babbitt, Ana Lucía García Briones, Ann Hayden, and David Festa**, “Better Access. Healthier Environment. Prosperous Communities. Recommended Reforms for the California Water Market,” Technical Report, Environmental Defense Fund 2016.
- Stavins, Robert N.**, “Transaction Costs and Tradeable Permits,” *Journal of Environmental Economics and Management*, 1995, 29, 133–148.
- SWRCB**, “Guide to Water Transfers in California,” Technical Report, State Water Resources Control Board, Division of Water Rights 1999.
- Tomori, Franceska, Erik Ansink, Harold Houba, Nick Hagerty, and Charles Bos**, “Market

power in California's water market," *American Journal of Agricultural Economics*, 2024, 106 (3), 1274–1299.

UNDP, *Human Development Report 2006: Beyond Scarcity: Power, Poverty and the Global Water Crisis*, New York: Palgrave MacMillan, January 2006.

Western Water Company, "Statement for the State Water Resources Control Board," Statement, Public Workshop on Water Transfers May 2000.

Williamson, Oliver E., "Transaction-Cost Economics: The Governance of Contractual Relations," *The Journal of Law and Economics*, October 1979, 22 (2), 233–261.

Womble, Philip and W. Michael Hanemann, "Water Markets, Water Courts, and Transaction Costs in Colorado," *Water Resources Research*, 2020, 56 (4), 1–28.

Wooldridge, Jeffrey M., *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., Cambridge, MA: MIT Press, 2010.

World Bank Group, "High and Dry: Climate Change, Water, and the Economy," Technical Report, World Bank Group May 2016.

Young, Robert A., "Why are there so few transactions among water users?," *American Journal of Agricultural Economics*, 1986, 68 (5), 1143–1151.

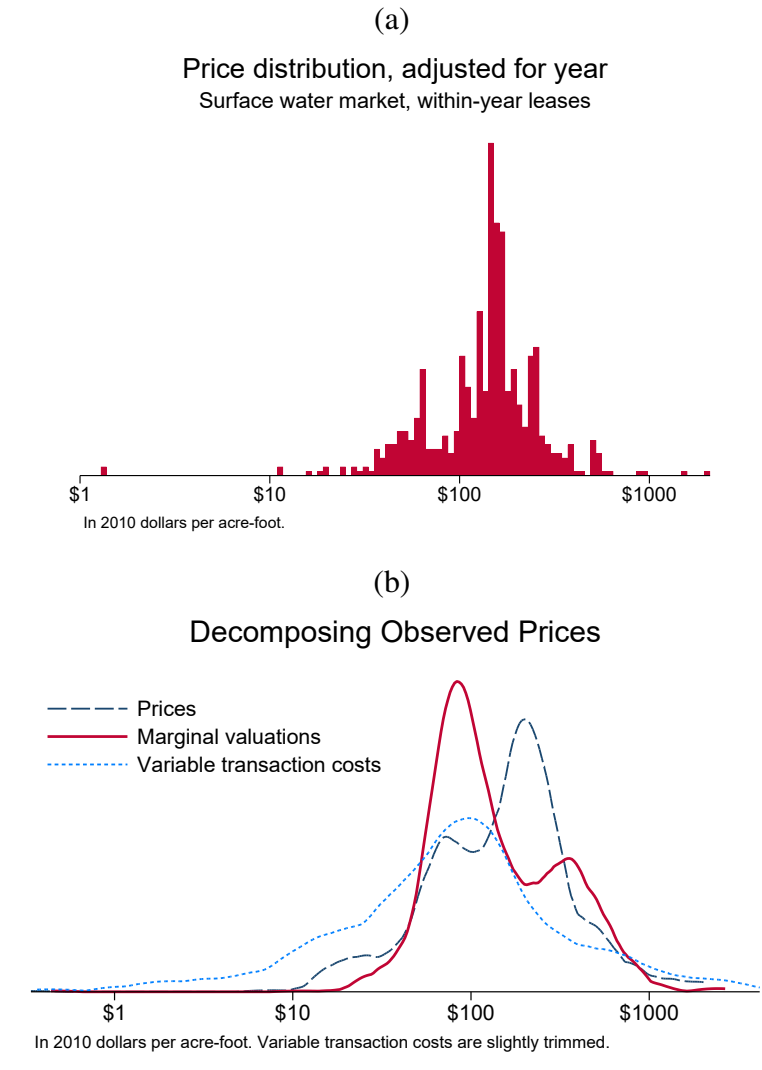


Figure 1: Panel (a): Distribution of surface water transaction prices in the data used for analysis, controlling for year, logarithmic scale. Observations are transactions of within-year leases of surface water observed in California, 1992-2015. The overall sample mean is added to the residual from a regression of log price on year fixed effects. Panel (b): Kernel density estimates of observed prices and estimated marginal valuations for the same transactions, logarithmic scale. Observations are surface water transactions made by each water district.

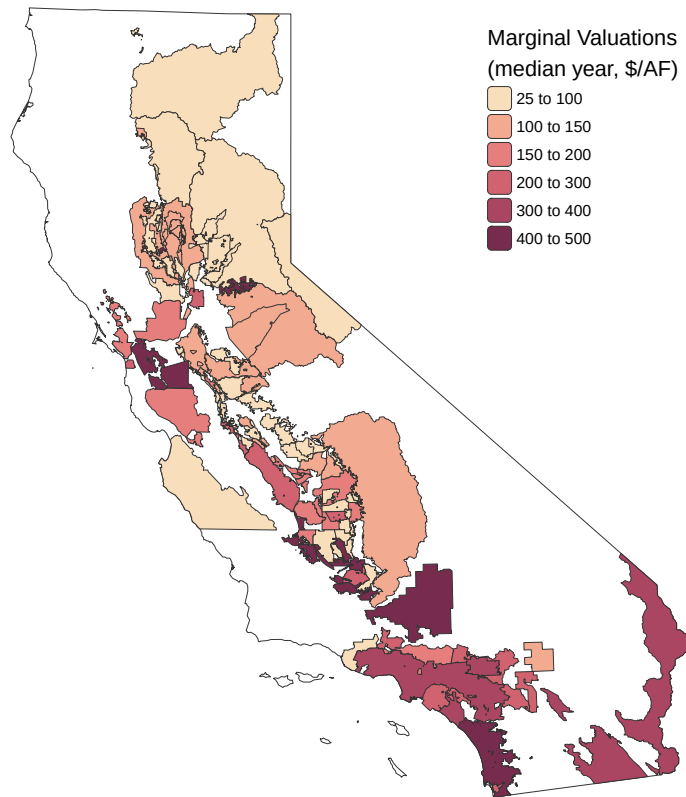


Figure 2: Mean estimated marginal valuations (per acre-foot) by water district for a median-year scenario. Estimates are obtained by evaluating the demand model at quantities observed in 2010. Boundaries are available for districts accounting for 89% of surface water in the simulation sample. Other districts are aggregated to planning areas, geographical regions defined by the California Department of Water Resources, taking means weighted by surface water quantity consumed.

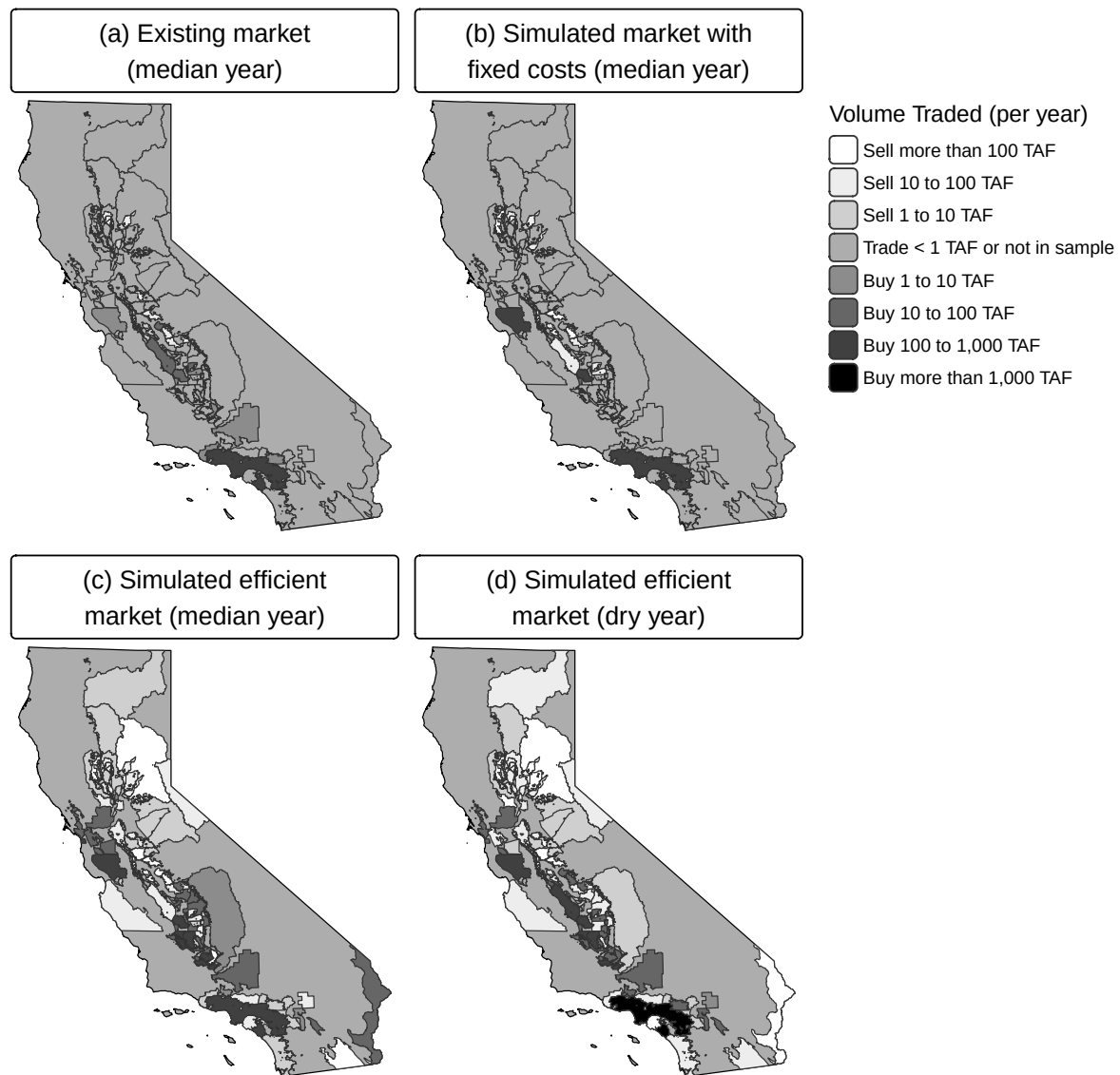


Figure 3: Net annual surface water quantities traded (TAF = thousand acre-feet) in observed transactions (Scenarios 1-2) and additional quantities traded in counterfactual simulations (Scenarios 3-4), by water district. The efficient market scenarios use the estimated demand model to maximize aggregate surplus, subject to physical conveyance costs calibrated from a hydrologic network model. The fixed cost scenario does the same, with trading restricted to districts predicted to enter the market. A color version of this figure is provided in Online Appendix Figure A4.

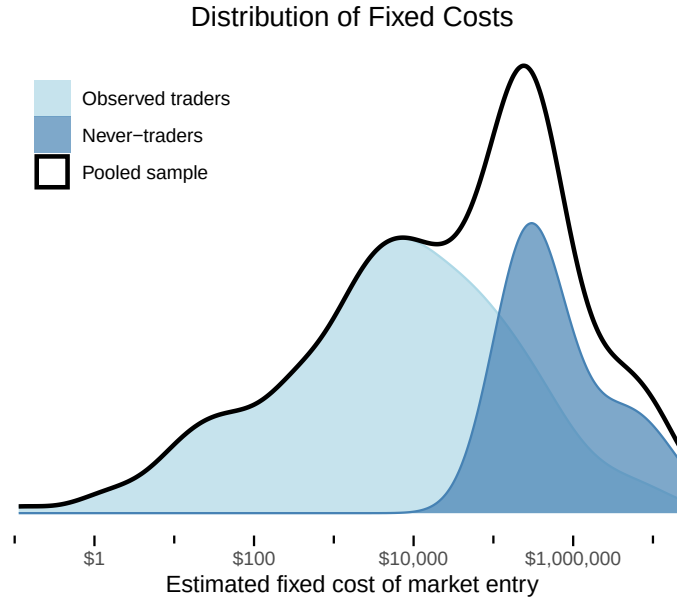


Figure 4: Kernel density estimates of estimated fixed transaction costs, logarithmic scale. Observations are water districts. “Observed traders” are districts ever observed in transactions data; fixed costs are estimated as the minimum surplus observed in any year of the data. “Never-traders” are districts included in the simulation sample but never observed in transactions data; fixed costs are estimated as the maximum surplus simulated in the efficient-market scenario. “Pooled sample” includes all districts from both samples.

Table 1: Transactions data

Panel A: Summary statistics					
Variable	Mean	SD	Obs.		
Volume (acre-feet)	9,571.7	28,224.7	1248		
Price (2010\$/acre-foot)	235.5	297.6	1259		
Distance conveyed (km)	206.1	269.7	1247		
Distance conveyed in rivers (km)	80.0	103.1	1247		
Distance conveyed in canals (km)	126.0	203.8	1247		
Distance of virtual conveyance (km)	86.4	118.5	1247		
Elevation lift (ft)	421.5	1,029.5	1247		
Crosses the Sacramento-San Joaquin Delta (=1)	0.27	0.444	1247		
Reviewed by the State Water Boards (=1)	0.411	0.492	1259		
Within project (=1)	0.422	0.494	1259		
Panel B: Transactions by sector					
	Sales		Purchases		Net Purchases
	Count	Volume (TAF)	Count	Volume (TAF)	Volume (TAF)
Agricultural	639	7,130.4	330	1,785.9	-5,344.5
Urban (Municipal & Industrial)	100	819.3	190	2,209.9	1,390.6
Panel C: Transactions by hydrologic region					
Central Coast	27	18.5	33	18.7	0.2
Colorado River	20	824.4	6	69.5	-755.0
North Coast	2	3.7	2	0.7	-2.9
Sacramento River	275	4,219.5	37	89.5	-4,130.1
San Francisco Bay	16	188.6	46	530.1	341.5
San Joaquin River	298	2,088.2	104	576.6	-1,511.6
South Coast	10	154.3	68	1,573.3	1,419.0
South Lahontan	2	14.4	30	11.1	-3.2
Tulare Lake	89	438.2	194	1,126.3	688.1

Dataset of spot-market transactions (i.e., within-year leases) of surface water in California made by each district (so transactions are repeated for each party). Panel A reports summary statistics of transaction characteristics. Panels B and C reports the count and total volume of transactions sold and purchased in each sector or hydrologic region. Net purchases within category do not sum to zero because inclusion criteria are applied separately to each side of a transaction (e.g., many agricultural sales are purchased by environmental or government entities, which do not appear in the table as buyers). TAF = thousand acre-feet.

Table 2: Variable Transaction Costs from Observed Determinants (Step 1)

	Dependent variable: Log Price					
	All determinants			Lasso-selected determinants		
	(1)	(2)	(3)	(4)	(5)	(6)
Seller × River dist. (km, 1000s)	−0.81 (0.76)	−0.84 (0.63)	0.22 (0.60)			0.45 (0.64)
Seller × Canal dist. (km, 1000s)	−0.31 (0.68)	−0.30 (0.73)	0.042 (0.80)			
Seller × Virtual dist. (km, 1000s)	−0.73 (0.47)	−0.68 (0.50)	−0.50 (0.44)			
Seller × Pumping lift (ft, 1000s)	0.025 (0.10)	0.075 (0.10)	0.027 (0.12)			
Seller × Delta crossing (=1)	1.0*** (0.26)	0.94*** (0.27)	0.42 (0.30)	0.66*** (0.14)	0.68*** (0.20)	0.31 (0.20)
Seller × State Boards review (=1)	0.53** (0.22)	0.51** (0.23)	−0.055 (0.27)	0.59** (0.26)	0.49* (0.26)	
Seller × Import into project (=1)	−0.0079 (0.064)	0.095 (0.091)	0.058 (0.059)			
Seller × Export from project (=1)	0.39*** (0.13)	0.28* (0.17)	0.35 (0.28)	0.31*** (0.096)	0.26* (0.14)	
Seller × Ag counterparty (=1)	0.0034 (0.096)	0.061 (0.11)	0.25*** (0.079)			0.20** (0.080)
−Buyer × River dist. (km, 1000s)	−0.91 (0.59)	−0.71 (0.53)	−0.65 (0.50)			
−Buyer × Canal dist. (km, 1000s)	−0.48 (0.66)	−0.15 (0.72)	0.25 (0.66)			
−Buyer × Virtual dist. (km, 1000s)	0.095 (0.36)	0.017 (0.49)	−0.23 (0.32)			
−Buyer × Pumping lift (ft, 1000s)	−0.20* (0.11)	−0.26* (0.14)	−0.36*** (0.13)			
−Buyer × Delta crossing (=1)	0.28 (0.21)	0.22 (0.16)	0.17 (0.17)			
−Buyer × State Boards review (=1)	0.31** (0.13)	0.31** (0.14)	0.30** (0.13)			0.17** (0.076)
−Buyer × Import into project (=1)	−0.31** (0.14)	−0.26** (0.13)	−0.13 (0.10)			
−Buyer × Export from project (=1)	0.56*** (0.20)	0.55*** (0.21)	0.54*** (0.19)		0.48 (0.31)	0.45** (0.22)
−Buyer × Ag counterparty (=1)	−0.20 (0.12)	−0.21* (0.12)	−0.16 (0.13)			
Side × District FE	✓	✓		✓	✓	
Side × Hydro. region × Year FE	✓			✓		
Side × Planning area × Year FE		✓			✓	
Side × District × Year FE			✓			✓
Observations	1,247	1,247	1,247	1,247	1,247	1,247
Observations excluding singletons	1,032	966	689	1,032	966	689
Clusters	113	105	76	113	105	76

Regressions of transaction price on observable cost determinants. Observations are transactions made by each water district, stacked such that each transaction is repeated for each district involved. Cost determinants are interacted with the district's side of the market (buyer or seller). On both sides of the market, positive coefficients indicate positive transaction costs. Seller-side coefficients compare prices across buyers, so a positive coefficient reflects a price premium. Buyer-side coefficients compare prices across sellers, but regressors are multiplied by -1 , so a positive coefficient reflects a price discount. California is divided into 10 hydrological regions and 56 planning areas. Standard errors in parentheses are clustered by transaction and district. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Market Entry and Inverse Price Elasticity of Demand (Steps 2-3)

	(1)	(2)	(3)	(4)
	Probit	First stage	Reduced form	IV Lasso
	Any Trades	Log Quantity	Adjusted Log Price	Adjusted Log Price
Log quantity consumed				-1.4** (0.46)
Log allocation %, own	0.021 (0.040)	0.16*** (0.045)	-0.11 (0.075)	
Log allocation %, rest of state	-1.9*** (0.44)	0.96*** (0.29)	-3.1*** (0.91)	
Inverse Mills ratio		-0.0030 (0.027)	-0.045 (0.061)	-0.092 (0.069)
Any trades, year $t - 1$	0.73*** (0.097)			
Any trades, year $t - 2$	0.92*** (0.12)			
Any trades, year $t - 3$	0.25*** (0.092)			
Any trades, sum of prior years	0.13*** (0.017)			
Log maximum water rights	0.18*** (0.030)			
Log cropland area	0.016 (0.027)			
Central Valley Project (=1)	0.50*** (0.077)			
State Water Project (=1)	0.89*** (0.15)			
Irrigation District (=1)	0.29*** (0.098)			
California Water District (=1)	0.30*** (0.10)			
Allocation variables (lasso-selected)	✓			
District fixed effects		✓	✓	✓
Linear time trend		✓	✓	✓
Observations	54,498	972	972	972
Clusters	2,355	155	155	155
Pseudo- R^2	0.59			
Lasso-selected instruments				3
First-stage F-statistic		13		46
Sup-score weak-ID test				reject

Observations are transactions made by each water district. Column (1) reports pooled probit estimates of the extensive-margin decision of market entry (i.e., the probability of any surface water transactions); the data also includes years when districts do not trade and districts that never trade. Columns (2) and (3) report first-stage and reduced-form effects of two allocation instruments on quantity consumed (of surface water) and marginal valuations (i.e, prices adjusted for variable transaction costs estimated in Step 1). Column (4) reports instrumental variable lasso estimates of the elasticity of marginal valuations with respect to quantity consumed, using the full set of candidate instruments. Columns (2)-(4) correct for sample selection using inverse Mills ratios from year-specific estimates of the model in column (1). Standard errors in parentheses are clustered by district in column (1) and calculated by Bayesian bootstrap with weights blocked by district in columns (2)-(4). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Annual Economic Gains from Surface Water Markets

Panel A: Observed Transactions in the Existing Market				
Conditions	Volume traded (TAF)	Average marginal valuation (\$/AF)	Gains (millions)	
Scenario 1. Observed market (spot market transactions)				
Dry	123	\$ 285	\$ 55.2	
Median	289	\$ 161	\$ 8.3	
Wet	35	\$ 129	\$ 1.5	
Panel B: Simulated Markets Without Transaction Costs				
Conditions	Additional volume traded (TAF)	Average equilibrium price (\$/AF)	Avoided costs of observed transactions (millions)	Gains from new transactions (millions)
Scenario 2: Efficient market (with only physical costs of conveyance)				
Dry	3,929	\$ 176	\$ 84.7	\$ 559.0
Median	3,338	\$ 137	\$ 45.1	\$ 211.3
Wet	3,502	\$ 126	\$ 4.6	\$ 220.3
Scenario 3: With fixed transaction costs				
Dry	1,823	\$ 238	\$ 71.1	\$ 393.1
Median	1,157	\$ 161	\$ 38.8	\$ 76.8
Wet	1,219	\$ 129	\$ 3.0	\$ 139.6
Scenario 4: No conveyance costs				
Dry	4,698	\$ 205	\$ 93.6	\$ 1,099.5
Median	4,982	\$ 165	\$ 82.0	\$ 635.2
Wet	5,283	\$ 151	\$ 11.0	\$ 721.0
Scenario 5: With a key environmental constraint (Sacramento River outflow held fixed)				
Dry	3,402	\$ 188	\$ 66.3	\$ 505.4
Median	3,016	\$ 139	\$ 43.3	\$ 199.1
Wet	3,251	\$ 127	\$ 4.2	\$ 213.7
Scenario 6: With more capacity constraints (Sac. River and Colorado River Aqueduct held fixed)				
Dry	3,325	\$ 189	\$ 66.3	\$ 503.0
Median	3,015	\$ 139	\$ 43.3	\$ 198.9
Wet	3,233	\$ 128	\$ 4.2	\$ 213.3
Scenario 7: Among only prior traders (districts observed in the transactions data)				
Dry	3,138	\$ 211	\$ 69.4	\$ 462.9
Median	2,340	\$ 154	\$ 43.8	\$ 132.7
Wet	2,701	\$ 126	\$ 3.3	\$ 177.6
Scenario 8: Linear demand (instead of isoelastic)				
Dry	2,658	\$ 133	\$ 80.4	\$ 360.5
Median	2,825	\$ 114	\$ 51.0	\$ 159.4
Wet	2,980	\$ 105	\$ 2.5	\$ 167.0

Per-year welfare analysis of the existing market (Panel A) and counterfactual simulations (Panel B). Each scenario is run under dry, median, and wet conditions, which draw quantities and trading volumes from the years 2014, 2010, and 2006, respectively. All dollar figures are in 2010 USD; gains are per year. TAF = thousand acre-feet.